

Recollection and familiarity in dense hippocampal amnesia: A case study

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Received 17 December 2004; received in revised form 16 May 2005; accepted 17 May 2005

Available online 14 July 2005

Abstract

In the amnesia literature, disagreement exists over whether anterograde amnesia involves recollective-based recognition processes and/or familiarity-based ones depending on whether the anatomical damage is restricted to the hippocampus or also involves adjacent areas, particularly the entorhinal and perirhinal cortices. So far, few patients with well documented anatomical lesions and detailed assessment of recollective and recognition performance have been described. We report a comprehensive neuroanatomical assessment and detailed investigation of the anterograde memory functions of a previously described severe amnesic patient (VC). The results of four previously published neuroradiological investigations (resting PET, qualitative MRIs, volumetric MRI and functional MRI) together with the results of two new investigations (voxel-based morphometry and magnetic resonance spectroscopy) are presented. The consistent finding across these different qualitative and quantitative examinations of VC's brain has shown that there is primarily structural and functional abnormality located selectively in the hippocampus bilaterally. Marked impairments in both verbal and non-verbal recall and recognition standardized memory tests were documented in the context of VC's intact cognitive profile and normal semantic memory. The results of five new experimental recognition memory tests tapping recollection and familiarity using verbal, topographical (buildings and landscapes) and unknown human faces memoranda revealed striking differential effects according to the type of stimuli used. A receiver operating characteristic analysis revealed that VC's recollective- and familiarity-based recognition processes were well preserved for unknown human faces. In contrast, recollective-based recognition for verbal and topographical material was at floor. Familiarity-based recognition was also impaired, significantly below controls for verbal and buildings memoranda and quite weak, although not reaching significance, for landscapes. These data suggest that the hippocampus is involved in recollective processes of verbal and topographical stimuli. It also plays an appreciable role in familiarity processes for these stimuli. However, recollection and familiarity of human faces appear not to depend on this region.

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Keywords: Hippocampus; Perirhinal cortex; Recognition memory; Amnesia; Recollection; Familiarity

1. Introduction

In the 1980's, several influential models of anterograde memory argued that anterograde memories can be retrieved through two processes: (1) recollection, where items are

recalled with associated phenomenological aspects of an encoded event and/or (2) familiarity, where stimuli are recognized without retrieval of other details (e.g. Jacoby & Dallas, 1981; Mandler, 1980; Tulving, 1985). It is widely accepted that normal recognition memory is subserved by both processes although some researchers proposed that familiarity-based processes alone subserve recognition memory adequately for, at least, some kinds of recognition task

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(e.g. Baddeley, Vargha-Khadem, & Mishkin, 2001; Brown & Aggleton, 2001; Mayes, Holdstock, Isaac, Hunkin, & Roberts, 2002).

Severe and global anterograde amnesia is usually observed following bilateral medial temporal lobe damage. Since Scoville and Milner (1957), many investigations have focussed on the role of the hippocampus (e.g. Gluck & Myers, 1993; Rudy & Sutherland, 1995). Recent research has suggested that areas subadjacent to the hippocampus, particularly the entorhinal and perirhinal cortices, may be crucial for at least some components of anterograde memory, particularly familiarity-based recognition memory (e.g. Aggleton & Brown, 1999; Mishkin, Suzuki, Gadian, & Vargha-Khadem, 1997). However, this position is controversial. Squire (1992) has influentially argued that familiarity-based recognition memory is part of a general declarative (explicit) memory system, and as such, is underpinned by the same anatomical network including the hippocampal formation and adjacent cortical areas (the entorhinal, perirhinal and parahippocampal cortices; Squire & Zola, 1997).

Important sources of evidence in this debate are studies of patients with selective lesions in the medial temporal lobe. However, there are very few patients with severe anterograde amnesia for which detailed neuroanatomical and neuropsychological data are available. The present study investigates one such patient. In the next section, we are going to review the lesion studies focusing on the anatomical structures involved in anterograde amnesia. We will review the contrasting evidence concerning whether the hippocampus plays a role on recollection only, or a role in both recollection and familiarity. It will be shown that the evidence appears inconsistent.

In recent reviews, Aggleton and Brown have supported the view that recognition is held to occur either through the recollection of the stimuli and/or by detecting stimulus familiarity (Aggleton & Brown, 1999; Brown & Aggleton, 2001). They suggested that the hippocampal system subserves recollective-based recognition processes. In contrast, the perirhinal cortex subserves familiarity-based recognition processes. In a meta-analysis of studies of anterograde amnesia that had used the recognition memory test (RMT; Warrington, 1984) they concluded that RMT scores were impaired when damage extended into other areas of the temporal lobes. In contrast, RMT scores were normal or near normal when damage was restricted to the hippocampus (Aggleton & Shaw, 1996). However, only three patients with assumed hippocampus damage were included in this analysis and two of them were impaired in at least one of the two subtests of the RMT (GD and AB; Shimamura & Squire, 1986).

A series of further studies also concluded that the hippocampus does not play a role in recognition memory. For example, Baxendale (1997) reported normal or near normal performance on the RMT in a group of patients with longstanding epilepsy and unilateral hippocampal sclerosis. However, longstanding epilepsy may be associated with atyp-

ical cerebral organization of cognitive functions. Turriziani, Fadda, Caltagirone, and Carlesimo (2004) reported a relative sparing of single item recognition in a subgroup of amnesic patients. However, it is difficult to interpret these findings. Quantitative neuroradiological data demonstrating the selective nature of their hippocampal damage was not provided. In addition their experimental tasks used only unfamiliar faces. It may well be that areas outside the hippocampus may subserve memory for this type of material (see below).

Holdstock, Gutnikov, Gaffan, and Mayes (2002), Holdstock et al. (2002) and Mayes et al. (2002) reported a patient (YR), who had a bilateral reduction of approximately 50% of the hippocampal volume. Despite being impaired on tests of recall, she performed in the normal range on almost all recognition memory tests. For example, YR was impaired on the recall subtests of the Doors and People test but was unimpaired on the recognition subtests (DPT; Baddeley, Emslie, & Nimmo-Smith, 1994). On recent investigation she was shown to be impaired in recall/recollection and recognition of associations between different kinds of information while showing preserved recognition for item, intra-items association and associations between items of the same kind. Additionally, her performance on remember/know tasks indicated a deficit in recollection while familiarity was spared, although the authors raised the possibility that she had some difficulty performing these tasks appropriately. The authors argued that YR's hippocampal lesion had selectively affected her recollective memory. Her spared familiarity-based recognition abilities were thought to be mediated by her intact medial temporal lobe cortices (Mayes et al., 2004). However, it should be noted that patient YR had an unusual and perplexing clinical presentation. She was found confused following an intravenous injection of opiate drugs. Following this episode she did not seek medical attention for her memory problems for a decade and she returned to independent living, including working as a clerk for several years. Such activities are usually difficult to carry out for acquired amnesic patients with hippocampal atrophy similar to YR (e.g. Manns & Squire, 1999; Rempel-Clower, Zola, Squire, & Amaral, 1996).

A pattern of memory profile similar to YR has been described in the context of developmental amnesia. Patient Jon had a long-standing reduction of 50% of hippocampal volume incurred perinatally which is equivalent to that of YR (patient Jon; Maguire, Vargha-Khadem, & Mishkin, 2001). Interestingly an fMRI study revealed that Jon's residual hippocampal tissue was active during memory retrieval (Maguire et al., 2001). Similarly to YR, Jon performed well on the recognition subtests of the DPT and poorly on the recall subtests. Jon's intact recognition memory performance was interpreted as mediated by unimpaired familiarity-based recognition judgements. Vargha-Khadem et al. (1997) suggested that perirhinal and entorhinal cortices are critical structures for such processes. However, it may well be that the effects of damage in the immature brain are quite different from those in the mature brain. Therefore, such conclusions

may not be easily generalizable (see also for further discussion Maguire, Frith, Rudge, & Cipolotti, 2005).

Recently, Mayes and co-workers have reported two important single case studies describing adult patients with hippocampal damage who show preserved familiarity and impaired recollection (Aggleton et al., 2005; Bastin et al., 2004). Bastin et al.'s (2004) patient (MR) performed well on several recognition memory tests despite being impaired in recall memory tests. His familiarity judgments were preserved while his recollection judgments were impaired. However, there are some puzzling aspects in this patient's performance. First, he showed fluctuations in his scores on a standard memory test such as the doors and the shapes subtests of the DPT. Secondly, the case description raised the possibility that he may have had some degree of psychiatric problems. Therefore, his recall memory impairment must be interpreted with caution. Thirdly, the patient produced a rather high number of false alarm rates in the task used to evaluate the contribution of familiarity and recollection (see Wixted & Squire, 2004b for discussion of the problems that false alarms create for the evaluation of familiarity and recollection). The patient (KN) described by Aggleton et al. (2005) showed deficits on standard tests of recall memory and selective sparing of familiarity in tests using remember/know and receiver operation procedures. However, this patient was also impaired on a standard visual recognition memory test and had a weak performance on a verbal recognition test given his high level of premorbid general ability. There was suggestion of involvement of other cortical and subcortical structures and on a previous volumetric analysis his hippocampi were reported as normal (Aggleton et al., 2005; McKenna & Gerhand, 2002). This evidence suggests that caution needs to be exerted before accepting the authors' conclusions that the "... hippocampus is vital for recall but not for recognition ...".

Yonelinas et al. carried out a series of interesting studies investigating recall and recognition memory in a group

of patients with mild hypoxia who were assumed to have hippocampal lesions on the basis of a relatively short coma duration (Yonelinas et al., 2002). Recall deficits were reported significantly greater than recognition deficits. This difference was thought to be due to a deficit in recollection but not familiarity. Using structural covariance modeling, recollection estimates were shown to correlate with the coma duration (a proxy measure of presumed hippocampal damage) while familiarity estimates did not. However, it should be noted that in five patients, the performance was equally impaired on the recall and the recognition task (see Fig. 1b; Yonelinas et al., 2002). This raises the possibility that hippocampal damage may also impair item recognition. In line with this, Wixted and Squire (2004a) have recently argued that, after excluding an outlier from the Yonelinas et al. (2002) control group, the data indicated that hippocampal damage affects to a similar extent recognition and recall. Yonelinas et al. (2004) did consider this artifact explanation and argued that it does not provide an adequate account for their results. However, regardless of whether the data are compatible or not with the view that recollection but not familiarity is decreased by hypoxia, the absence of neuroradiological data in their study must make any assertion about selective hippocampal damage uncertain (for a similar view, see also Wixted & Squire, 2004b).

Turning to discuss the evidence suggesting that the hippocampus plays a role in both recollection and familiarity, Chan, Revesz, and Rudge (2002) have reported neuropathological findings on patient NT, who became densely amnesic following a unilateral right temporal lobectomy for epilepsy which included the hippocampus, the uncus and the amygdala (Warrington & Duchon, 1992). The patient presented with severe global impairment both in recognition and recall memory tests. Her postmortem pathological re-examination revealed sclerosis of the unresected left hippocampus but no significant pathology extended beyond the hippocampus.

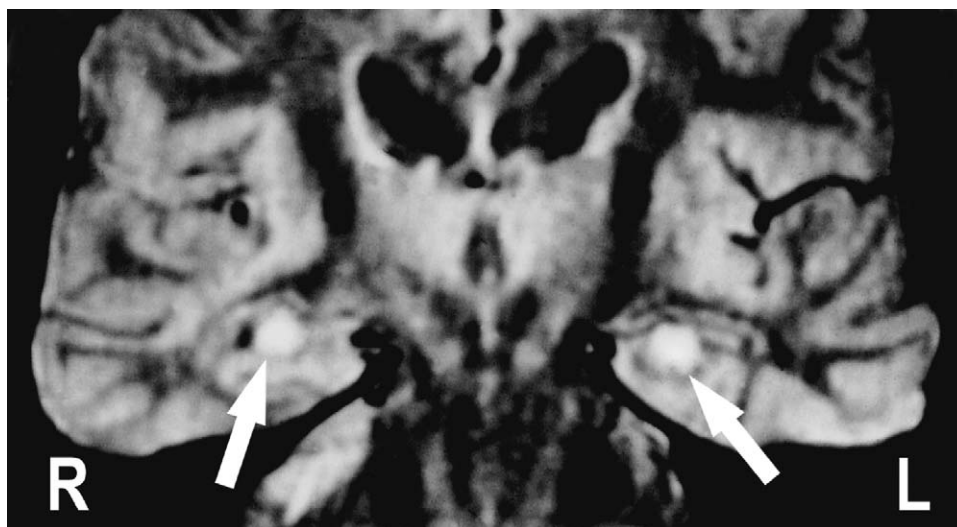


Fig. 1. Coronal sections through temporal lobe at the level of the body of the hippocampus. High signal return seen in each hippocampus (TR 2000/TE30 ms and 5 mm thick).

The appearance of the resected right temporal lobe was unremarkable. Thus, it is tempting to conclude that NT's recognition memory deficit was caused by the bilateral hippocampal damage.

Squire and co-workers have provided direct evidence that damage to the hippocampus may cause comparable impairments in recognition and recall (e.g. Squire & Zola, 1997). They have described at least 12 patients with hippocampus damage who were impaired on a range of recognition memory tests (e.g. Manns & Squire, 1999; Manns, Hopkins, Reed, Kitchener, & Squire, 2003; Reed & Squire, 1997). Moreover, in two further studies, they showed that recognition memory, including the component that supports familiarity judgments, depends on the integrity of the hippocampus (Knowlton & Squire, 1995; Manns et al., 2003). However, we noted that some patients had damage extending outside the hippocampal formation (GD, LM and WH; Rempel-Clower et al., 1996); measurements for cortical areas were not available for others (Manns et al., 2003) and only qualitative or no MRI information was available for some cases (PH, LJ; Reed & Squire, 1997; AB, Manns et al., 2003; AB, Reed & Squire, 1997). Therefore, this anatomical evidence is not conclusive.

This review of the lesion studies, focusing on the anatomical structures concerning the anatomical underpinnings of recollection and familiarity processes in anterograde memory indicate that the available evidence is inconclusive. Indeed, we concur with Mayes et al.'s view that "... more patients with apparently selective hippocampal lesions urgently need to be identified and given extensive hypothesis-driven neuropsychological assessments ..." (Mayes et al., 2002, page 339).

We report a further examination of the profoundly amnesic patient VC (Kartsounis, Rudge, & Stevens, 1995) with an extensive and ungraded retrograde amnesia. Previous MRI volumetric measurements revealed gross abnormalities in both hippocampi which were markedly shrunk (Cipolotti et al., 2001). A recent fMRI study revealed that his residual hippocampal tissue did not show changes in activity during memory retrieval; the patient was effectively ahippocampal (Maguire et al., 2005). In this current investigation, we undertook detailed assessment of his anterograde amnesia. We evaluated recollective- and familiarity-based recognition processes for three different types of memoranda: verbal, topographical and human faces. Two further neuroanatomical investigations were also undertaken (voxel-based morphometry and magnetic resonance spectroscopy). The results contribute to the debate on the role of the hippocampus and related structures in recollection and familiarity.

2. Case description

The patient (VC), 74-year-old at time of this investigation (born 1926), is a retired chief engineer in large ships, such as liners, who was reported by his wife as having an excellent memory. In May 1992, he developed an apparent severe

migraine attack followed by a seizure. In September 1993, he had two further seizures 4 days apart with a tachyarrhythmia requiring cardioversion. Following these episodes, at the age of 67, he was left profoundly amnesic. Since then he had no further epileptic episodes. Neurological examination revealed a profound amnesia and a minor but variable impairment of pain assessed by pin prick over the left hand and foot. A mild impairment of left ventricular function of echocardiography was also found (see for details, Cipolotti et al., 2001).

3. Neuroradiological findings

We are going to review first the results of four neuroradiological investigations which have been previously published (resting PET, qualitative MRIs, volumetric MRI and functional MRI) and secondly we are going to report the results of two new investigations (voxel-based morphometry and magnetic resonance spectroscopy).

3.1. Resting PET

Resting PET scans have been previously obtained (see Kapur, Thompson, Kartsounis, & Abbott, 1999). Decreased tracer uptake was seen throughout the right thalamus and possibly the right parietal region compared to the left, in addition to the hippocampal lesion.

3.2. MRI findings

T2-weighted images of the whole brain were obtained in the axial, coronal and sagittal planes (Signa 1.5 T MRI System, GE, Milwaukee) (Cipolotti et al., 2001). Increased signal return was found throughout the length of both hippocampi (see Fig. 1). Formal measurement of the T2 relaxation time was in excess of 90 ms (greater than 3 standard deviations (S.D.) above normal) at all levels of both hippocampi. The hippocampi were atrophied and there was also abnormal signal return from the left amygdala. There was no evidence of abnormal signal return from any other part of the brain. In particular, both thalami and parietal lobes were normal (Cipolotti et al., 2001). A structural MRI scan acquired contemporaneously with the fMRI study confirmed the selective bilateral hippocampal damage (Maguire et al., 2005).

3.3. Volumetric MRI findings

Previous detailed MRI volumetric measurements (based on region of interest, ROI, metrics) revealed gross abnormalities in both hippocampi which were markedly shrunk (left hippocampus between 5 and 6S.D.; right hippocampus between 4 and 5S.D. below the control mean) (Cipolotti et al., 2001). The grave hippocampal volume loss was present throughout the length of both hippocampi. In contrast, the volumes of the grey matter of both entorhinal cortices were entirely normal. Indeed, in our control sample of six age-

matched healthy control males there were individuals with entorhinal cortex volumes smaller than VC, who did not show any obvious memory problems. Similarly, the volumes of both right and left temporal lobes were within normal limits as were the volumes of the temporal gyri when measured individually. There were no obvious morphological abnormalities of the grey matter of the parahippocampal gyrus, but the white matter layer was noted to be reduced in size, particularly at the rostral end of the gyrus between the head of the hippocampus superiorly and the entorhinal cortex inferiorly (see for further discussion [Cipolotti et al., 2001](#)). The volume of the right parahippocampal gyrus was less than 1S.D. below the control mean, while the volume of the left parahippocampal gyrus was slightly reduced (between 2 and 3S.D. below the control mean). The difference between the two sides was not significant. This mild degree of volume loss in the left parahippocampal gyrus was clearly much less than that noted for the hippocampi and was attributed to the reduced size of the white matter layer.

3.4. Functional MRI findings

We conducted a fMRI study when VC was successfully retrieving basic autobiographical facts and general knowledge ([Maguire et al., 2005](#)). The paradigm used activated in controls the hippocampus and a largely medial and left lateralized network of brain regions. Interestingly, patient Jon with bilateral anoxic hippocampal damage of perinatal origin activated the hippocampus to a similar extent to controls but, at difference, a bilateral brain network ([Maguire et al., 2001](#)). VC's hippocampus did not show any changes in activity during memory retrieval (autobiographical facts and general knowledge). This was different from both the controls and patient Jon. However, his memory network was found to be largely medial and left lateralized (medial frontal, retrosplenial, temporoparietal junction, lateral temporal cortices, temporal pole and the left parahippocampal gyrus). This was identical to healthy controls but quite different from patient Jon. Of particular interest is the result indicating that VC was activating the left parahippocampal gyrus, given the previous findings of slight reduction of volume in this area. A direct comparison showed that it was more active in control subjects. It could be that this finding of reduced, but importantly not absent, activation merely reflects the overall slight volume reduction rather than grey matter dysfunction.

3.5. Voxel-based morphometry (VBM)

Our previous qualitative MRI analyses revealed that there were no other areas of damage, apart for the hippocampi and left amygdala abnormality. In particular, the small fraction of the perirhinal cortex contained within the parahippocampal gyrus, namely the portion located at the fundus of the collateral sulcus, appeared qualitatively intact, as did the remainder of the perirhinal cortex ([Cipolotti et al., 2001](#)). The MRI volumetric analysis confirmed the marked and selective hip-

pocampal damage. The fMRI study demonstrated that VC was effectively ahippocampal. However, he was still able to activate the left parahippocampus and a largely medial and left lateralized neuronal network.

We also employed two additional methods of quantitative analysis. The first, VBM is a relatively recent development in brain imaging that allows whole brain structural images to be analyzed in an automated manner without the specification of any *a priori* regions of interest. It is a spatially specific unbiased method of analysing MR images reflecting macroscopic regional differences. It eschews observer bias inherent in region of interest metrics. It detects regional structural changes in both grey and white matter on a voxel by voxel basis between groups of subjects ([Good et al., 2001b](#); [Wright et al., 1995](#)).

Our aim was to use VBM analysis to assess the whole brain MRI scans of VC, and in particular, the structural integrity of his medial temporal lobes. This analysis allows us to cross-validate our previous findings based on ROI techniques measurements. A study comparing VBM with the ROI measurements, previously used in our volumetric MRI analysis indicated that the two techniques, although with some differences, detect a similar trend of atrophy in Alzheimer and semantic dementia patients ([Good et al., 2002](#)).

3.5.1. Controls

Fifteen healthy male subjects mean age 73 years (range, 69–79) were used as normal elderly control group.

3.5.2. MRI scanning

High resolution volumetric MR imaging was performed on a Siemens 2 T Magnetom scanner using an optimized MPRAGE sequence which affords enhanced grey/white matter contrast and segmentation ([Deichmann, Good, Josephs, Ashburner, & Turner, 2000](#)). The acquisition parameters included: TR/TE/TI 11/4/1000, flip angle 12, matrix 256 × 224, FOV 256 mm × 224 mm; 176 sagittal slices, 1 mm isotropic voxels.

3.5.3. VBM analysis

An optimized method of VBM was used. This method has been described in detail elsewhere ([Ashburner & Friston, 2000](#); [Good et al., 2001a,b](#)). Briefly this involves a number of fully automated pre-processing steps; including extraction of brain, spatial normalization into stereotactic space, segmentation into grey and white matter and CSF compartments, modulation for volume changes induced by spatial normalization and smoothing with a 10 mm full width at half maximum (FWHM) isotropic Gaussian kernel. After smoothing, each voxel represents the local average amount of grey (or white) matter in the surrounding region, the size of which is defined by the size of the smoothing kernel. We used a customized grey matter template for spatial normalization, which was created from a separate group of 15 elderly male subjects aged 65–80 years. This involved normalizing all the structural scans to the SPM T1 template, segmenting each

normalized image smoothing each grey matter segment with an 8 mm FWHM smoothing kernel. Finally, all the smoothed segments were averaged to create a grey matter template.

3.5.4. VBM statistical analysis

The smoothed grey and white matter segments were analysed using MATLAB 5.3 (MathWorks, Natick, MA, USA) and SPM99 (Wellcome Department of Imaging Neuroscience, ION, London) employing the framework of the general linear model. Regionally specific structural differences were assessed statistically using a two-tailed test, namely testing for increases or decreases in grey (and white) matter. Significance levels were set at $P < 0.05$, corrected for multiple comparisons.

3.5.5. Results

When correcting for comparisons across the whole brain, VBM detected no atrophy in any areas. However, this is not unusual, given the power of VBM to detect atrophy in a single case (Mehta, Grabowski, Trivedi, & Damasio, 2003). Given the results of our previous volumetric analysis, we therefore investigated which areas showed atrophy at uncorrected thresholds (using a $P = 0.001$ threshold). Despite wide inter-subject variability in medial temporal lobe structures in our elderly control group, VBM detected grey matter atrophy in the head and body of both the hippocampi, particularly medially. This atrophy was significant on the right and just failed to reach significance on the left (right hippocampus, $P = 0.001$, $t = 3.8$; left hippocampus, $P = 0.002$, $t = 4.37$; see Figs. 2 and 3). Grey matter atrophy also extended into the upper margins of the entorhinal cortex bilaterally although this too failed to reach significance (right entorhinal cortex upper border, $t = 2.71$, $P = 0.004$; left entorhinal cortex upper border, $t = 2.82$, $P = 0.004$; see Fig. 3). Importantly, there were no grey matter differences in the bulk of the entorhinal cortex between VC and controls while they were apparent for

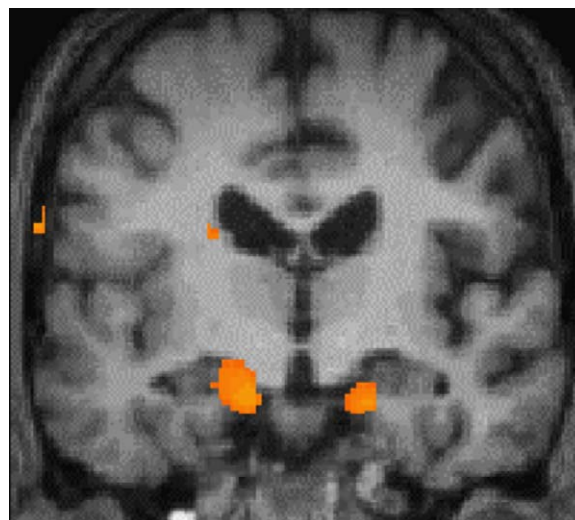


Fig. 3. VBM detected atrophy in the medial aspects of both hippocampi, bordering on the upper margin of entorhinal cortex bilaterally (the images are presented at the threshold of $P < 0.001$ for display purposes and clarity. The significant results are discussed in the text).

the hippocampus (see Figs. 4 and 2). Furthermore, no grey matter atrophy was detected in perirhinal cortex, in the temporal lobes or elsewhere in the cortex. The only area of white matter atrophy detected was adjacent to the lateral border of the right hippocampus ($t = 4.76$, $P < 0.0001$; see Fig. 5). This corroborates previous independent ROI-based measurements (Cipolotti et al., 2001).

3.6. Magnetic resonance spectroscopy findings

The second new means of analysis was magnetic resonance spectroscopy. Single voxel proton magnetic resonance spectroscopy (^1H MRS), carried out by means of a GE Sigma 1.5 T scanner at an echo time (T_E) of 30 ms and repetition time (T_R) of 2000 ms, using the PRESS sequence to localize

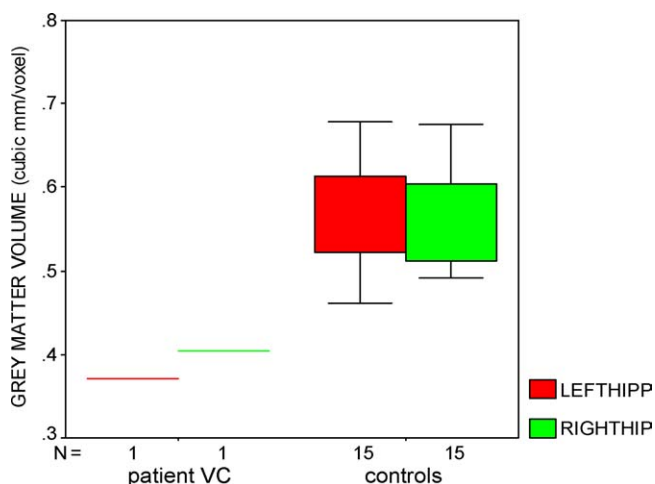


Fig. 2. Box and whisker plot of the absolute amount of grey matter within a single voxel (mm^3 of grey matter in a mm^3 voxel) placed within the hippocampus heads showing a significant difference between VC and controls.

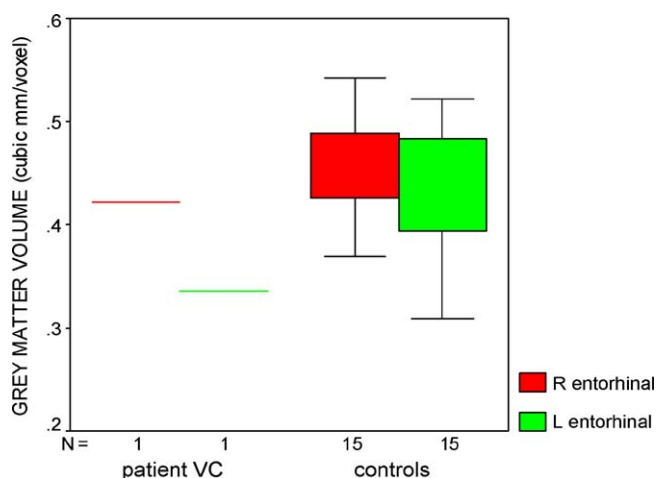


Fig. 4. Box and whisker plot of the absolute amount of grey matter within a single voxel (mm^3 of grey matter in a mm^3 voxel) placed within the entorhinal cortex showing no difference between VC and controls.

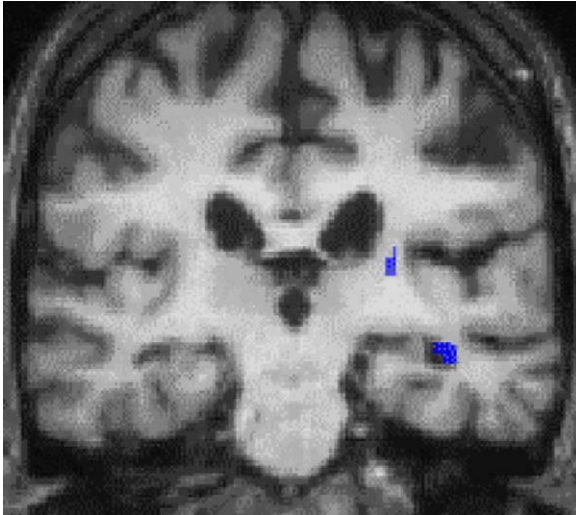


Fig. 5. VBM detected white matter atrophy adjacent to the lateral margin of the right hippocampus (the images are presented at the threshold of $P < 0.001$ for display purposes and clarity. The significant results are discussed in the text).

the volume of interest. The spatial resolution for this procedure depends on voxel size, which in the present study varied between 1.5 and 4.9 ml. In general, the larger the voxel the better the signal-to-noise ratio, but volumes are constrained by the topographical properties of the region being examined. In the present case, for instance, the acquisition of spectra from the hippocampal formation alone required that the voxel did not overlap with any of the surrounding areas. Spectra were shimmed on the water signal, resulting in full bandwidth at half maximum (FWHM) values of 2 Hz for the medial and temporal–occipital regions. For the frontal, thalami and hippocampi, the FWHM values were 5, 4 and 6 Hz, respectively.

Identical areas from left and right cerebral hemispheres were studied in the frontal and medial and lateral temporo-occipital regions, thalami and the hippocampi. Concentrations of *N*-acetylaspartate (NAA) were derived using the linear combination (LC) model (Provencher, 1993) and expressed both as absolute concentrations and as ratios to creatine (Cre). NAA is found almost exclusively in healthy neurons, and can therefore be taken as an index of neuronal integrity or function. Creatine concentration is unaffected by pathological change, and is therefore commonly used as a reference metabolite. NAA values were compared with values obtained from normal controls (using similar parameters, but on a different MRI scanner).

3.6.1. Results

Values of NAA and NAA:Cre from the patient's hippocampi were considerably lower than those obtained from control subjects. In sharp contrast, the values for areas outside the hippocampus were normal, symmetrical and similar to those obtained from the normal control subjects. In addition, they were consistent with those reported in the literature.

Thus, the measurements of neuronal integrity in VC's frontal, medial and lateral temporo-occipital regions and the thalami were normal.

In summary, the consistent findings across several and different qualitative and quantitative examinations of the patients' brain show that there is primarily structural and functional abnormality located selectively in the hippocampi bilaterally.

4. General cognitive tasks

The patient was first referred to the Neuropsychology Department of the National Hospital for Neurology and Neurosurgery in September 1993 and was subsequently reassessed on four occasions. The results of his first four assessments, together with experimental investigations are reported elsewhere (Cipolotti et al., 2001; Kartsounis et al., 1995, 2001). The experimental investigation that will be described here took place around the time of his fifth and last neuropsychological assessment (April 2000). The results of his five neuropsychological assessments are reported in Table 1.

At the time of his fifth neuropsychological assessment his verbal IQ remained in the upper end of the average range while his performance IQ had further improved and was at a very superior level. His performance on the graded difficulty naming test (GNT, McKenna & Warrington, 1983) remained at a high average level and his performance on the object decision test of visuoperception remained at a normal level (Warrington & James, 1991). On three tests of frontal 'executive' skills (cognitive estimates, Shallice & Evans, 1978; Wisconsin card sorting, Nelson, 1976; Hayling test, Burgess & Shallice, 1997) his performance was entirely satisfactory. In particular, on the Wisconsin card sorting test he obtained the six categories rapidly and made no perseverative errors. His attention and concentration abilities were clearly completely intact. The most notable feature at this time remained his global amnesia.

In summary, VC presented with an intact performance on general intelligence, language, perception and frontal 'executive' tasks which remained static over a 7-year period. This indicated that there was no progressive cognitive decline. An in depth assessment of his severe memory impairment is documented below.

5. Semantic memory tasks

In our previous investigation of patient VC we documented an impairment in new semantic learning (Cipolotti et al., 2001). A recent review suggested that the role of the hippocampus in semantic memory remains unclear and advocated more systematic studies (Spiers, Maguire, & Burgess, 2001). Traditionally, semantic memory tests are thought to tap the integrity of the functioning of the temporal lobes.

Table 1
Cognitive test scores

	March 1993	March 1996	February 1997	April 1998	April 2000
Verbal IQ	99	105	102	105	107
Performance IQ	111	120	136	141	138
GNT (O)	20/30 (25–50%ile)	17/30 (10–25%ile)	23/30 (75%ile)	23/30 (75%ile)	24/30 (75–90%ile)
GNT (P)	n.t.	n.t.	20/30 (75%ile)	n.t.	n.t.
Incomplete pictures	20/20	20/20	n.t.	n.t.	n.t.
Cube analysis	10/10	10/10	n.t.	n.t.	n.t.
Object decision	n.t.	18/20	17/20	17/20	18/20
Weigl	Passed	Passed	Passed	n.t.	n.t.
cognitive estimates	Passed	Passed	Passed	n.t.	Passed
Wisconsin card sorting	n.t.	n.t.	Passed	Passed	Passed
Hayling test	n.t.	n.t.	n.t.	17 (average)	18 (average)

GNT (O), graded difficulty naming test—objects; GNT (P), graded difficulty naming test—proper nouns; %ile, percentile; n.t., not tested.

We administered to VC a battery of tests designed to assess input to and output from central representation of knowledge via different sensory modalities. This battery comprises tests of naming, category fluency and comprehension. The results obtained are reported in Table 2.

5.1. Picture naming

Two tests of picture naming were administered; the Cambridge new naming test and the category-specific names test (Bozeat, Lambon Ralph, Patterson, Garrard, & Hodges, 2000;

McKenna, 1998). The Cambridge new naming test comprises 64 pictures of 32 living and man-made objects. The 32 living items comprises domestic animals, foreign animals, birds and fruit (eight examples for each category). The 32 man-made objects comprises large household items, small household items, vehicles and tools. The category-specific names test comprises realistic color photographs of items belonging to four semantic categories: animals, fruit, small manipulable and large objects. For each category there are 30 pictures. This test is graded in difficulty.

VC's performance was almost flawless on the Cambridge new naming test and well within normal limits in the category-specific names test.

Table 2
Semantic memory test scores (April 2000)

	No. correct
Cambridge new naming test	63/64
Living items	31/32
Man-made objects	32/32
Category-specific naming test	
Animals	23/30
Fruit	18/30
Small manipulable objects	20/30
Large objects	20/30
Category fluency (1 min)	
Animals	14
Birds	10
Dogs	7
Household	16
Vehicles	13
Tools	8
Fruits	10
Boats	8
F	15
A	18
S	17
Graded synonyms test	46/50 (75%ile)
Camel and cactus test	
Words	62/64
Pictures	61/64
Cambridge comprehension test	
Living	32/32
Non-living	32/32

%ile, percentile.

5.2. Category fluency

There were eight main semantic categories (animals, birds, dogs, household, vehicles, tools, boats and fruit) and three phonemic categories (F, A and S). One minute per category was allowed. His performance was also normal on both semantic and phonemic category fluency tasks.

5.3. Comprehension

Three tests of comprehension were administered. The graded synonyms test (Warrington, McKenna, & Orpwood, 1998), the camel and cactus test (pictures and words, Bozeat et al., 2000) and the Cambridge comprehension category test (Hodges & Patterson, 1995). VC's performance on all three tasks was again entirely satisfactory.

In summary, VC had an intact performance on all the semantic memory tasks administered. This indicated that his semantic memory is preserved.

6. Recognition and recall memory tasks

Throughout the five neuropsychological investigations, VC was assessed on a variety of anterograde memory tests involving both recognition and recall paradigms (see Table 3). Verbal and visual recognition memory was assessed using

Table 3

Anterograde memory test scores (No. correct)

	March 1993	March 1996	February 1997	April 1998	April 2000
Recognition memory words	33/50 (<5%ile)	35/50 (5%ile)	35/50 (5%ile)	36/50 (<10%ile)	35/50 (5%ile)
Recognition memory faces	32/50 (<5%ile)	34/50 (<5%ile)	41/50 (25–50%ile)	39/50 (=25%ile)	n.t.
Topographical memory test	13/30 (=5%ile)	n.t.	14/30 (5–10%ile)	13/30 (=5%ile)	12/30 (<5%ile)
Story recall					
Immediate	5 (<5%ile)	n.t.	n.t.	7.5 (<5%ile)	6 (<5%ile)
Delay	0	n.t.	n.t.	0	0
List learning			18/95 (<5%ile)	n.t.	
Paired associates T1	n.t.	n.t.	n.t.	4/24 (5%ile)	3/24 (<5%ile)
T2	n.t.	n.t.	n.t.	8/24 (5%ile)	7/24 (<5%ile)
Rey–Osterrieth figure					
Copy	10/36 (<10%ile)	n.t.	n.t.	35/36 (90%ile)	35/36 (90%ile)
Delay	0/36	n.t.	n.t.	5/36 (<5%ile)	3/36 (<5%ile)

%ile, percentile; n.t., not tested.

alternative versions of the RMT and on the topographical recognition memory test (Warrington, 1984, 1996). Verbal recall memory was assessed on the story recall and the list learning subtests of the adult memory and information processing battery (Coughlan & Hollows, 1985) and on the paired associates learning test. Visual recall memory was assessed on the Rey–Osterrieth complex figure. His performance was globally and severely impaired. The only exception being an improvement on the visual version of the RMT which employs unknown human faces. Whether this indicates a relative sparing of recognition memory for faces will be investigated in more detail below.

At the time of his fifth neuropsychological investigation (April, 2000) a further formal reassessment of his memory functions was undertaken. His verbal and topographical recognition memory skills remained severely impaired. He obtained a score at the 5th percentile on the verbal version of the RMT and below the fifth percentile on the topographical memory test. His recognition memory for faces was not reassessed on this occasion. However, it was noticeable that on the previous two assessments (February 1997 and April 1998) his scores were at or above the 25th percentile. When asked to recall the Coughlan and Hollows; Story and the Rey–Osterrieth figure, following a 30-min delay, he remained unable to remember even being exposed to them (Coughlan & Hollows, 1985; Osterrieth, 1944). Similarly, his performance was very impaired on the paired associated learning test (Warrington, 1996).

In summary, the results of the anterograde memory assessment indicate the presence of a severe global memory impairment affecting both recognition and recall which remained static over a 7-year period.

6.1. The Doors and People task

This task was administered, according to the published manual, at the time of his 4th and 5th neuropsychological assessments (Baddeley et al., 1994). It comprises four sub-

tests matched for difficulty, two tapping verbal and visual recognition memory and two tapping verbal and visual recall memory. The two recognition subtests are: verbal recognition: names test; visual recognition: doors test. The two recall subtests are verbal recall: names test; visual recall: shapes test. In the recognition subtests, there were 24 names on the verbal version and 24 doors on the visual version. On the recall subtests there were four pictures on the verbal version and four simple line drawings on the visual version. Recognition memory was tested with a four forced-choice format. Recall memory was tested with an approximately 8-min delay.

Analysis of VC's performance on the various subtests revealed that he had a grave and global memory impairment. He did not achieve the cut-off for an overall-scaled score of 1 (see Table 4).

Interestingly, this marked memory impairment equally affected recognition and recall processes. Indeed he performed systematically below the 1st percentile both on the two recognition subtests as well as the two recall subtests on both assessments.

Table 4

Doors and Peoples test scores (No. correct)

	April 1998	April 2000
Door and peoples test		
Names test		
Immediate verbal recall	6/36 (<1%ile)	7/36 (<1%ile)
Delayed verbal recall	3/12	4/12
Verbal recognition test A	5/12 (<1%ile)	4/12 (<1%ile)
Verbal recognition test B	2/12 (<1%ile)	3/12 (<1%ile)
Doors visual recognition test		
Test A	7/12 (<1%ile)	7/12 (<1%ile)
Test B	1/12 (<1%ile)	3/12 (<1%ile)
Shapes test		
Immediate visual recall	13/36 (<1%ile)	12/36 (<1%ile)
Delayed visual recall	4/12	4/12

%ile, percentile.

Table 5
VC and controls' performance on the five new Recognition Memory tests: analysis 1

Test	VC	Controls, mean (S.D.)	Number of controls	<i>T</i>	Probability
Words	0.65	1.65 (0.27)	15	−3.54	<0.005
Buildings	0.41	1.61 (0.44)	12	−2.59	<0.05
Landscapes	0.54	1.72(0.50)	12	−2.25	<0.05
Faces	1.55	1.73 (0.51)	9	−0.33	0.374

The results on the two word recognition memory tests are combined.

7. Five new verbal and non-verbal recognition memory tests and receiver operating characteristics analysis (ROCs)

The results obtained in the previous tests clearly indicated that VC presented with severe and global anterograde memory impairment affecting both recall and recognition memory tests. In the following series of recognition memory tasks we investigated the relative contribution of recollection and familiarity to VC's residual recognition skills. We devised five recognition memory tests: two verbal tests and three visual tests. Following previous studies we adopted an ROC analysis of recognition memory (Ratcliff, Sheu, & Gronlund, 1992; Yonelinas, Krollm, Dobbins, Lazzara, & Knight, 1998). To estimate recollection and familiarity we plotted ROCs for recognition judgements made on a 6-point confidence scale.

7.1. Controls

Twenty-three elderly adults (19 males, 4 females) served as controls. Their mean age was 76.3 years old (S.D. = 2.7, range = 70–80). Their NART estimated IQ was 110.6 (S.D. = 9.8, range = 96–129). All subjects were neurologically intact. Not all controls were administered all five memory tests. However, it was ensured that the subsample that served as controls for each test were representative of the larger "pool" in terms of both age, NART IQ and sex distribution. The number of controls for each individual test is presented together with the results (see Table 5).

7.2. Materials

The stimuli used were words, unknown human faces, unknown buildings and unknown landscapes. For each of the two verbal recognition memory tests, there were two sets (A and B) each of 160 words from the Toronto word pool, comprising both concrete and abstract words. The stimulus words were typed in upper case letters onto a white card. For each of the three visual recognition memory tests there were 160 black and white photos. The stimuli used for the faces test were unknown Caucasian faces with no obvious distinguishing feature. The stimuli used for the buildings test were unknown buildings with no obvious distinguishing features and with verbal cues digitally removed (e.g. house numbers, street names, etc.). The stimuli used for the landscapes

test were unknown landscapes representing a selection of different natural scenes with no obvious distinguishing features (e.g. stimuli such as buildings, people, animals, etc. were absent).

7.3. Procedure

The same procedure was adopted for all five recognition memory tests. There was a study phase and a test phase. The study phase required the subjects to decide whether (1) the word was "concrete" or "abstract" (words test); (2) the face was "pleasant" or "unpleasant" (faces test); (3) the architecture was "pleasant" or "unpleasant" (buildings test); or (4) the subject would like/not like to visit the location pictured (landscapes test). Study items were presented individually at an interval of 2–3 s.

In the test phase, the study items and lure items were individually, randomly intermixed presented. Subjects made recognition judgments after each item. Judgments were made according to a 6-point confidence scale. One corresponded to the most confident judgement that the stimulus was "new". Six corresponded to the most confident judgement that the stimulus was "old" ("new" and "old" referred to previously unstudied and studied stimulus, respectively). It was stressed that participants had to make full use of the 6-point scale by spreading their responses across all the possible confidence ratings.

7.4. Design

Pilot studies suggested that verbal recognition memory tests tended to be slightly easier than visual tests. This is in keeping with standard findings (e.g. Sweet, Demakis, Ricker, & Millis, 2000). In order approximately to match for difficulty the verbal and visual memory tests, we used a slightly different design. This was to ensure that a task difficulty artifact could not differentially affect performance.

For each of the 2 verbal tests, there were 80 stimuli in the study phase and 160 stimuli in the test phase. The 80 stimuli of the study phase were presented all in one block, immediately followed by the 160 stimuli used in the tests phase. Similarly, for each of the 3 visual tests, there were also 80 stimuli in the study phase and 160 stimuli in the test phase. However, to decrease the cognitive load, we presented the stimuli into two blocks separated by a short break (no less than 10 min). In each block in the study phase, we presented 40 stimuli and 80 stimuli in the test phase. In our subsequent analyses, we sum across the results obtained in the two blocks for each visual memory test (total score = 160 test stimuli). Despite the relatively large number of study items, this was no indication that VC or any of the controls suffered from a lack of attention or motivation during the tasks.

7.5. Analyses

We conducted two separate analyses. The first analysis aimed to assess the overall performance of VC and the con-

trols on the five tests. Therefore, the distinctions among the different confidence levels (1–3 and 4–6) were collapsed to calculate the proportion of hits and false alarms to studied and unstudied items, respectively. The sensitivity of each subject was calculated as d' -prime (d') using standard signal detection theory methodology (see Macmillan & Creelman, 2003). The performance of VC was compared to that of the control group using the procedure of Crawford and Garthwaite (2002).

The second analysis investigated the contribution of familiarity and recollection to the recognition judgements made by subjects. The ROCs of VC and the controls were plotted in Excel. We planned to fit them using the sum of squares search algorithm used by Yonelinas et al. in previous studies (e.g. Yonelinas et al., 1998). This procedure makes the assumption that recognition reflects the contribution of two independent processes recollection and familiarity (the dual process theory). However, using this algorithm we obtained clearly unsatisfactory estimates of recollection in some of our elderly controls. A few of them performed very well (approximately 95% hits) by using a confidence level which gives in no more than 50% false positives. At the same time, probably typically for elderly adults, when using the most stringent confidence level, the same controls obtained hits only of the order of approximately 30–40%. Application of the Yonelinas algorithm led to very high d' values for familiarity (with an extreme value of 2.65) together with impossibly low – because negative – recollection parameters (for the corresponding case –2.09). Indeed, of the recollection values estimated from the elderly control subjects, 7/48 had negative recollection parameter values, and these tended to be found with very high values of familiarity. A procedure was therefore adopted for estimating recollection using the hit and false alarm rates for the most conservative criterion adopted by a subject. This adopted estimate (hits–false alarms) gives a reasonable approximation to the tangent of the ROC curve where it intersects the y-axis (see Yonelinas et al., 1998). Familiarity estimates were then obtained by constraining the intercept with the y-axis according to the probability that an item was recollected (see above) and then fitting the points of the ROC curve using a standard equal variance model. This was carried out using a least-squares method with the solver in Excel, using the program developed by Yonelinas et al. (1998). This procedure preserves the main assumptions of the dual process theory. However, it avoids the problem of impossibly low recollection estimates. By using this modified procedure, we obtained satisfactory fits to the ROC curves of both VC's and all of the individual controls. The performance of VC was again compared to that of the control group using the procedure of Crawford and Garthwaite (2002).

7.6. Results

The two word tests gave virtually identical results both in VC and the controls. Therefore, we summed across the results of these two tests to give only one verbal memory result. In contrast, interesting differences were found in the perfor-

Table 6

VC and controls' performance on the five new recognition memory tests: analysis 2

	VC	Controls mean (S.D.)	<i>T</i>	Probability
Recollection				
Words	0	0.38 (0.14)	–2.46	<0.05
Buildings	0.025	0.44 (0.09)	–4.48	<0.001
Landscapes	0.05	0.44 (0.13)	–2.99	<0.01
Faces	0.21	0.36 (0.18)	–0.77	0.231
Familiarity				
Words	0.63	1.24 (0.32)	–1.81	<0.05
Buildings	0.45	1.19 (0.38)	–1.87	<0.05
Landscapes	0.43	1.18 (0.43)	–1.66	0.06
Faces	1.28	1.27 (0.36)	+0.02	0.491

The results on the two word recognition memory tests are combined.

mance of the visual memory tests. Therefore, these results are reported separately for each of the three visual memory test.

The results of the first analysis summarizing the overall performance of VC and controls and indicating their sensitivity on each of the memory tests are shown in Table 5.

VC was significantly impaired on the two verbal recognition memory tests. Similarly, his performance was also significantly impaired on the two memory tests employing topographical memoranda (buildings and landscapes). It should be noted that VC's performance on the verbal and topographical recognition memory tests, although markedly impaired, was nevertheless above chance. Strikingly, his performance was normal in the recognition memory test using unfamiliar faces. These results confirm previous findings, indicating that VC's recognition memory is severely impaired. However, they also suggest the remarkable and selective preservation of a particular type of non-verbal memory dedicated to faces. It is unclear to what extent recollective-based and familiarity-based processes are contributing to VC's impaired verbal and topographical recognition memory. The analysis of the contribution of these two processes in his spared recognition memory for human faces is also of interest. The results of the second analysis, investigating the contribution of familiarity and recollection to recognition are shown in Table 6.

VC's recollection processes for both verbal and topographical (buildings and landscapes) memoranda were at floor. Indeed, his recollection estimates were significantly impaired when compared with the control sample. In sharp contrast, his recollection processes for unfamiliar faces were preserved. It is sometimes held that normal subjects of VC's age have little recollective experience for material like faces (Bastin & Van der Linden, 2003). However, our analysis indicated that our elderly sample used recollection like responses well above chance. With faces all our elderly control subjects used the maximum rating of '6' significantly more often for old than for new stimuli. Strikingly, these included also VC (18 '6' responses for old stimuli and 1 '6' response to new stimuli). VC was at the 45th percentile in his frequency of use of '6' responses (old–new). By comparison, he was far below the worse control in use of '6' responses for the words, buildings and landscapes.

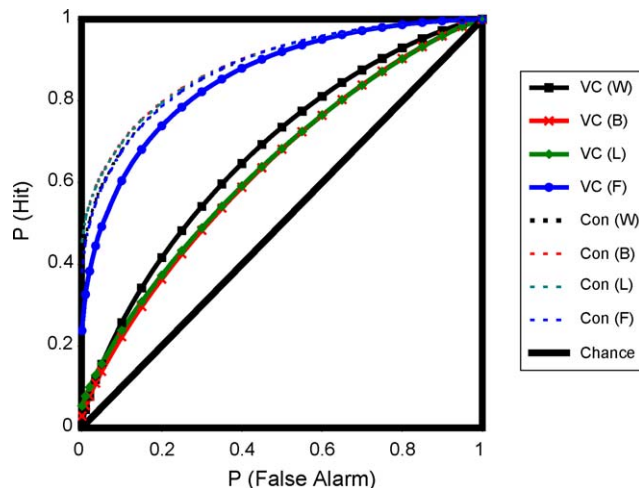


Fig. 6. VC and control's performance on the new recognition memory tests.

VC's familiarity processes were also very impaired. His performance was significantly poorer both for the verbal and the buildings memoranda. For landscapes, his estimates of familiarity were also poor. Indeed this deficit was also close to statistical significance. The failure to reach significance may well be due to the large variability in the control sample on this particular test. Remarkably, his familiarity processes for human faces were also normal. Indeed, familiarity estimates were indistinguishable from those of the controls. This interesting pattern of results is illustrated in Fig. 6. The proportion of new items accepted as old and the proportion of old items accepted as old are plotted on the x - and y -axis, respectively.

Through inspection of Fig. 6 it is clear that VC's ROCs for verbal and topographical memoranda are symmetrical along the diagonal. In contrast, his ROC curve for faces is clearly asymmetric, indicating the contribution of recollection processes. Similarly, the controls' ROC curves are also asymmetrical for all types of material.

VC's ROCs for verbal and topographical memoranda are also less curved than both his ROC for faces as well as the controls' ROC curves. This indicates that familiarity for verbal and topographical memoranda is greatly attenuated in VC. In contrast, his familiarity processes for faces are preserved.

In summary, our data suggest that VC's recollection-based processes were virtually eliminated for verbal and topographical memoranda. His residual above chance recognition memory performance for these memoranda appeared to be driven solely by rather marginally spared familiarity-based processes. In contrast, both VC's recollection and familiarity processes were entirely spared for human faces.

8. Discussion

The role of the hippocampus in anterograde memory, in particular in recall and recognition, remains a key issue in the cognitive neuroscience of memory. It is well recognized

that studies combining detailed neuroradiological and neuropsychological investigations in patients with anterograde amnesia and restricted anatomical lesions are of great theoretical interest. In this series of experiments, we have focussed on the anterograde component of patient VC's global amnesia. Detailed investigation of his recall and recognition memory performance was undertaken. Particular attention was given to the relative contribution of recollection and familiarity in his residual recognition memory for three different types of memoranda: verbal, topographical (buildings and landscapes) and unknown human faces. Anatomical evidence from both qualitative and quantitative neuroradiological investigations was also gathered. This was to attempt to clarify the structural integrity of the hippocampus and related anatomical structures with the view to then inferring their roles in the anterograde mnemonic processes.

As far as the extent of VC's anatomical lesion is concerned we have first reviewed four independent neuroradiological investigations performed in previous studies (resting PET, qualitative MRIs, MRI volumetry and fMRI). In addition, we report the findings from two new investigations (VBM and spectroscopy). A resting FGD PET scan showed, in addition to the hippocampal lesion, less signal return from the right thalamus and possibly the right parietal region (Kapur et al., 1999). However, there was neither analogous sign of structural abnormality on the two qualitative MRI scan examinations nor, more critically, of neuronal dysfunction of the thalamus on the magnetic resonance spectroscopy. Even if one was to interpret the isolated PET finding on the right thalamus as suggesting that this structure is dysfunctional, this can hardly explain VC's global memory impairment (for relevant unilateral thalamic patients, see Speedie & Heilman, 1982, 1983 and for further discussion, see Cipolotti et al., 2001).

Previous qualitative examinations of VC MRI studies documented damage to both hippocampi, with significant bilateral loss of volume and abnormal signal return (Cipolotti et al., 2001; Maguire et al., 2005). Abnormal signal return was also found in the left amygdala (for discussion of the relevance of the left amygdala damage, see Cipolotti et al., 2001). There was no evidence of abnormal signal return from any other part of the brain, including the anterior temporal lobes and the aforementioned thalamus.

The results from our previous quantitative anatomical study of VC's brain based on region of interest analysis were in good accordance with these findings (Cipolotti et al., 2001). Gross abnormalities were revealed in both hippocampi, which were markedly shrunk throughout the entire length. The volume of the entorhinal cortex was normal. Similarly, the volume of the temporal lobes and their gyri were normal. The right parahippocampal gyrus was also normal. A slight volume reduction was found only in the left parahippocampal gyrus. This appeared to be due to a slight reduction in the size of the white matter layer, rather than morphological abnormalities in the grey matter (Cipolotti et al., 2001). Interestingly, in our functional MRI study we

found that this area was active in VC during memory retrieval. Direct comparison showed that it was more active in control subjects (Maguire et al., 2005). It could be that our finding of reduced, but importantly not absent, activation in this region in VC merely reflected the overall slight volume reduction rather than grey matter dysfunction.

The results of the fMRI study revealed that VC is functionally amnesic. On tasks in which he was able to successfully retrieve autobiographical fact and general knowledge, he did not activate the hippocampus, unlike elderly controls. However, in common with controls he activated medial and lateral regions such as the medial frontal, retrosplenial and lateral cortices, temporal pole, temporo-parietal junction and the aforementioned left parahippocampal gyrus (Maguire et al., 2005).

Two additional quantitative neuroradiological investigations were conducted in this study. We first used VBM to examine whole brain MRIs scans paying particular attention to medial temporal lobe structures. By using VBM we aimed to investigate macroscopic abnormalities in an unbiased fashion in order to gather evidence to add to our previously reported findings of volumetric abnormalities assessed using an observer dependent technique. So, the main relevance of our VBM findings is in the context of the other neuroradiological findings. This technique has been criticized (e.g. Bookstein, 2001; Crum, Griffin, Hill, & Hawkes, 2003) and it has been regarded of limited use in single case studies (e.g. Mehta et al., 2003; Salmond et al., 2002). However, it has been recognized that VBM provides results similar to those of volumetric MRI (e.g. Good et al., 2001a,b) and has the potential to reveal areas of damage that may be missed using expert tracing of lesions (e.g. Mehta et al., 2003). VBM automated analysis detected grey matter atrophy in both the head and the body of the hippocampus which extended into the upper border of the entorhinal cortex. However, no significant atrophy was detected in the entorhinal cortex itself. The perirhinal cortex, which was not assessed by our previous ROI imaging technique, was found to be normal. Similarly, neither the temporal lobes nor the rest of the cerebral cortex were significantly different in VC compared with the elderly controls. It does remain possible that small white matter abnormalities, which are thought to be common in the ageing brain, have gone undetected because of VBM's poor spatial resolution power. However, our control subjects were sex and age matched to VC. Moreover, on T2 scans no abnormalities were detected either in the elderly controls or on VC, apart from his known hippocampal damage.

Secondly, we have used magnetic resonance spectroscopy analysis to evaluate neuronal integrity in VC's brain. This analysis revealed that the only areas of abnormality were the hippocampi. Of particular note were the normal measurements of neuronal integrity in VC's frontal, medial and lateral temporo-occipital regions and in the thalami. Thus, the results of both VBM and magnetic resonance spectroscopy analyses have provided converging evidence with our previous qualitative MRIs, volumetric MRI and fMRI findings. Although

one cannot rule out the possibility of damage elsewhere in the brain of VC, our findings consistently suggested that in his brain the primary abnormality is located in the hippocampi bilaterally.

The performance of VC on a comprehensive range of cognitive tests is entirely in keeping with the neuroradiological results. VC showed intact performance on tests of intelligence, frontal 'executive', perceptual and language functions over a 7-year period. This suggests that not only he did not have progressive cognitive decline but also that cognitive modules requiring frontal, occipital, parietal and temporal areas are functioning normally. Most importantly, his performance was entirely normal on a wide variety of stringent semantic memory tests. Both neuroimaging and lesion studies have indicated that semantic memory is underpinned by a neuronal network based upon temporal lobe structures (e.g. Vandenberghe, Price, Wise, Josephs, & Frackowiak, 1996). Patients with semantic dementia, who typically show a severe impairment on tasks, such as those administered to VC, present with clear cut damage in the temporal, entorhinal and perirhinal cortices (e.g. Chan et al., 2001; Mummery et al., 2000). The completely normal performance of VC on stringent semantic memory tests supports the conclusion that his extra-hippocampal temporal areas are functioning normally. The only cognitive impairment we could document in VC was his severe amnesia. Given that the detailed neuropsychological evaluation demonstrated the selective nature of his memory impairment and the comprehensive neuroradiological evaluation demonstrated damage primarily located to the hippocampus, we consider that his episodic memory impairment is related to this lesion.

Our investigation of the anterograde aspects of VC's global amnesia revealed a severe deficit encompassing both verbal and non-verbal memory. Interestingly, not only was VC's anterograde amnesia severe, but also it was found to equally compromise recognition and recall memory on a series of standardized anterograde memory tests. Of particular relevance is the performance of VC on two subsequent administrations of the Doors and People test which comprise verbal and visual recognition and recall tests matched for difficulty (Baddeley et al., 1994). On both administrations of this test, VC failed to attain the cut-off for an age-scaled score of 1 in the recognition and recall verbal and visual subtests.

VC's performance on five new anterograde recognition memory tests designed to investigate the relative contribution of recollection and familiarity revealed yet again a strikingly severe recognition impairment, with the exception of one remarkable preservation. Thus, VC's performance on two verbal memory tests was impaired. Similarly, his performance on two visual memory tests for topographical memoranda (e.g. buildings and landscapes) was also impaired. By contrast his performance on the visual memory test requiring the recognition of unfamiliar faces was spared. Interestingly, his scores on the RMT for faces were less impaired than his scores on the RMT for words and for topography.

Often impairments of memory for topographical memoranda occur together with impairments for face memory (e.g. Clarke, Assal, & de Tribolet, 1993; Landis, Cummings, Benson, & Palmer, 1986; McCarthy, Evans, & Hodges, 1996; Pallis, 1955). However, there are a few single cases described in the literature with a highly selective visual memory impairment. So, for example, we described impaired recognition memory for unfamiliar faces in the context of normal recognition memory for unknown buildings and landscapes (Blair, Frith, Smith, Abell, & Cipolotti, 2002; Cipolotti & Maguire, 2003; Cipolotti, Robinson, Blair, & Frith, 1999; Maguire & Cipolotti, 1998). The opposite pattern of impairment, namely intact memory for unfamiliar faces but impaired topographical memory, is also on record (e.g. Carlesimo, Fadda, Turriziani, Tomaiuolo, & Caltagirone, 2001; Incisa della, Cipolotti, & Warrington, 1996). These findings suggest that there may be different types of visual memory which are subserved by different neuronal structures. Indeed a number of neuroimaging and animal studies have suggested that processing of faces and landmarks is underpinned by separate neuronal networks (e.g. Aguirre, Zarahn, & D'Esposito, 1998; Buckley, Booth, Rolls, & Gaffan, 2001; Epstein & Kanwisher, 1998; Epstein, Harris, Stanley, & Kanwisher, 1999; Kanwisher, McDermott, & Chun, 1997; Maguire, Mumery, & Buchel, 2000; Wada & Yamamoto, 2001). Recent neuropsychological findings on seven patients with focal brain injury, suggest that the hippocampus may be involved in the processing of spatial scenes (landscapes) but not in the processing of faces (Lee et al., 2005). This would fit with VC's hippocampal damage being responsible for his impaired recognition memory for topographical memoranda. This finding is especially important since impairment on tests of topographical memory would be difficult to explain in terms of sub-optimal functioning of the left parahippocampal gyrus. His spared recognition memory performance for faces would also provide further evidence for normal functioning of extrahippocampal areas. A number of extrahippocampal areas have been identified as playing a role in face processing. In particular, the fusiform gyrus has been identified as a key area (e.g. Kanwisher et al., 1997; Maguire, Frith, & Cipolotti, 2001). We would like to speculate that perhaps this area may underpin VC's spared recollection and familiarity for faces.

Receiver operating characteristics analyses, designed to investigate the relative contribution of recollection and familiarity, were carried out on the performance of both VC and age matched controls subjects on the five new anterograde recognition memory tests. There is a debate as to whether the dual process model is the optimal way of fitting ROCs (Glanzer, Kim, Hilford, & Adams, 1999). However, in our study we found that the ROCs of both VC and the controls were well fitted by the dual process model. This suggests that our estimates of recollection and familiarity gives reliable values of the contribution of these two processes. Analysis of VC's confidence judgements revealed that recollection-based processes were completely at floor both for words and topograph-

ical memoranda. In striking contrast, recollection-based processes were spared for unfamiliar faces. Familiarity-based processes were also impaired for both verbal and topographical memoranda. These findings are consistent with the results obtained in the standardized recognition memory tests used and therefore provide a degree of cross-validation of these analyses. Conversely, VC's familiarity estimates for unfamiliar faces were indistinguishable from the controls. Altogether these results suggest that VC had a severe impairment in recollection judgments for words and topographical memoranda, but not for faces. Familiarity judgments for verbal and topographical memoranda were also impaired. Since VC's recollection was at floor, but his recognition memory for these memoranda was above chance, it seems likely that his impaired familiarity processes were underpinning the recognition performance.

We suggest that our findings support the view that the hippocampus subserves processing (or learning) that is not available to related extrahippocampal areas or the neocortex. This processing appears to be critical for the effective remembering of new events related to verbal and topographical memoranda. According to this and in keeping with our findings, the hippocampus not only must be critical for recollective, but also play a role in familiarity-based processing for these memoranda. In this context, we noted that in a series of influential studies by Squire and co-workers found no sparing of recognition ability relative to recall (Reed & Squire, 1997; Zola-Morgan & Squire, 1986). For example, Manns et al. (2003) investigated recollection and familiarity-based recognition in seven patients with bilateral damage thought, on indirect evidence, to be limited primarily to the hippocampal region. The authors used eight recognition memory tests: four verbal and four non-verbal tests. Two of the non-verbal tests used abstract drawings and two used unfamiliar faces. Unfortunately the authors collapsed the data across all four non-verbal recognition memory test, so it is impossible to ascertain whether the patients' performance differed in the tests using unfamiliar faces. However, the results indicated that all patients were equally impaired on both recollection and familiarity-based recognition. Interestingly, Reed and Squire (1997) reported the effect of bilateral hippocampus damage on recognition memory for words and faces. Almost all the patients were unimpaired on an immediate test of face recognition, similar to that used in this study. Altogether this evidence supports the view that the hippocampus is critical not only for subserving recollection mnemonic processes but also plays an important role in familiarity processes. We concur with this view and further suggest that the hippocampus is critical for both recollection and familiarity for verbal and topographical memoranda. However, recollection and familiarity of human faces appears not to depend on the hippocampus.

Our results on patient VC may be considered not easily reconcilable with fMRI studies of human memory in normal subjects reporting greater activity in the hippocampus during recollective recognition than in tasks more likely to depend on

familiarity (Davachi, Mitchell, and Wagner, 2003; Eldridge, Knowlton, Furmanski, Bookheimer, & Engel, 2000; Mitchell & Wagner, 2003; Ranganath et al., 2004). However, it should be noted that the hippocampal damage in VC completely abolished recollection while it led to impaired, but above chance, familiarity. This suggests that the hippocampus is completely critical for recollection and also that it contributes, although not exclusively, to familiarity. Thus, our findings are compatible with the fMRI findings of greater activation of the hippocampus during a recollection response than a familiarity one. Moreover, some fMRI studies of the role of the hippocampus in familiarity combine the effects of familiarity within the confidence levels for an overall Yes response and an overall No response (e.g. Yonelinas, Otten, Shaw, & Rugg, 2005). Thus, if the hippocampus is also involved in the encoding of new information (Rugg, Fletcher, Frith, Frackowiak, & Dolan, 1997; Tulving, Markowitsch, Kapur, Habib, & Houle, 1994) in this type of functional imaging paradigm the role of the hippocampus in processing familiarity for previously seen items becomes difficult to disentangle from responses to novel stimulus. Indeed in such a paradigm normal subjects may be immediately aware if a particularly vivid new stimulus (e.g. the word *explosion*) was not previously presented. This is because they know confidently that they would have remembered it. This will contribute to the ‘familiarity’ measure. However, this process is not available to VC.

Of course our investigation cannot irrefutably eliminate the possibility of hidden cortical pathology i.e. damage undetectable through the use of current neuroimaging techniques (e.g. Aggleton & Brown, 1999; Bayley, Hopkins, & Squire, 2003; Markowitsch et al., 1997). For example, it has been suggested that hypoxic injury following a cardiac arrest, as suffered by VC, may result in generalized cerebral atrophy and involve aspects of cognition other than memory (e.g. Caine & Watson, 2000; Grubb et al., 2000). However, using currently available imaging techniques we have been unable to detect any meaningful damage/dysfunction in VC’s brain outside the hippocampus. The only exception is the slight decrease of the left parahippocampal volume. However, this structure was shown to activate on the fMRI study and any abnormality here can hardly account for VC’s impairment on topographical memory. Moreover, using detailed cognitive testing we have been unable to identify any clinically meaningful cognitive deficits other than episodic memory. It remains of course possible that undetected extrahippocampal pathology, perhaps due to the etiology of VC’s cerebral damage or to his aging brain, may underpin his impaired familiarity processes for verbal and topographical material. However, VC’s intact cognitive profile and his selective sparing of both recollection and familiarity for unfamiliar faces are perhaps surprising, if one was to accept that he had additional extra-hippocampal damage/dysfunction.

We wish to stress that the majority of papers reporting amnesic patients tend to identify ‘selective hippocampus damage’ on the basis of either visual ratings of structural

qualitative MRI scans or usage of only one quantitative neuro-radiological technique. Often comprehensive behavioral data are not available. In our study, we have attempted to obtain data using a variety of different and independent quantitative neuroradiological and cognitive investigations. To our knowledge no other amnesic patient has been so extensively studied using such a variety of different investigations. Rosenbaum et al. (2005) suggested that a viable approach to the investigation of memory functions in brain damaged patients would require putting equal emphasis on both the tests used to identify the pattern of impaired and spared functions as well as on the size and location of the lesion itself. In this respect, our investigation has attempted to meet this requirement and has produced results indicating that the hippocampus makes essential contributions to recognition memory.

However, we are aware of the important differences between VC’s anterograde memory profile and that of well studied acquired amnesic patients such as, for example, YR (e.g. Holdstock et al., 2002; Mayes et al., 2004), MR (Bastin et al., 2004), KN (Aggleton et al., 2005) and the developmental amnesic patient Jon (e.g. Vargha-Khadem et al., 2003), who were described as having intact recognition memory. However, with the exception of Jon none of the above adult acquired amnesic patients were as impaired as VC and the patients of Squire and co-workers on memory tests. Therefore, at present, it remains difficult to make direct comparisons with these patients. Moreover, it remains an open question for future research to further the understanding of the factors which may play an important role for the differences in the patients’ memory profiles. Certainly factors, such as, for example, etiology of lesion, age and individual differences need to be considered further. The possible functionality of residual hippocampal tissues and the possibility of brain’s functional reorganization also need to be mentioned. Thus, our fMRI results in VC with adult-acquired anoxic hippocampal pathology (Maguire et al., 2005) were quite different from those obtained in Jon who has sustained anoxic hippocampal damage perinatally (Maguire et al., 2001). These results suggest that the age at which hippocampal damage occurs may have implications for the subsequent functionality of residual hippocampal tissue with a possible knock-on effect on the wider memory network (for further discussion, see Maguire et al., 2005). Perinatal hippocampal damage appeared to result in some preserved functionality in residual hippocampal tissue and lead to more widespread reorganization involving homologous areas in both hemisphere. In contrast, damage later in life, as in the case of VC, appears to result in a lack of hippocampal activity and much more circumscribed functional plasticity. Neuroimaging studies investigating acquired patients whose memory profile differ from VC are clearly needed in order to explain differences and similarities between different patients.

In summary, we have documented a detailed anatomical and cognitive investigation of a patient with restricted anatomical lesions and severe memory impairment. Our findings suggest that the hippocampus appears to be involved both

in recollection- and familiarity-based recognition judgments for verbal and topographical memoranda. In contrast, recollection and familiarity for unfamiliar faces seemed not to be impaired by hippocampus damage. Since this is the first study to highlight such a dissociation we are currently investigating further this issue in patients with selective hippocampal pathology.

Acknowledgements

We thank Dr. Peter Garrard, Dr. Eleanor Maguire and Prof. Andy Yonelinas for their help and advice in the manuscript preparation. Ms. Hannah Heims kindly collected controls' data. Chris Bird was supported by a MRC Cooperative Programme Grant No. G9900106. Lastly, we are tremendously grateful to VC for participating in this study.

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